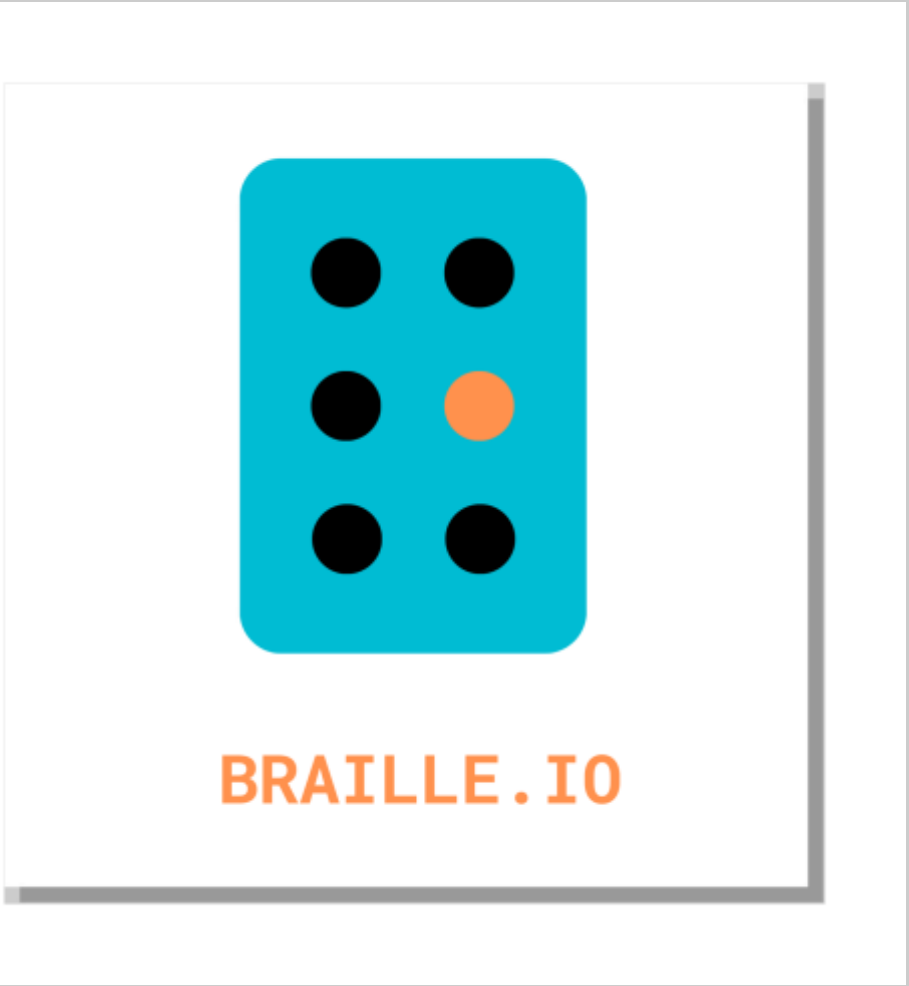

BRAILLE.IO

Details:

- **STUDENT'S NAME-**

1)Pratik Bhaskar Patil

2) Manthan Namdev Oulkar

Problem definition:



Visually impaired individuals face significant challenges in learning Braille due to the lack of accessible, interactive, and beginner-friendly learning tools. Traditional methods are often slow, dependent on trainers, and fail to provide real-time feedback, making the process difficult for complete beginners.

Our Solution:

Braill.io:



Working of Braille.io:

Mode I:

In Mode 1, the device provides a user-friendly interface with two buttons, one for advancing through the alphabet and the other for moving backward.

When the user presses either button, the device generates Braille alphabets and verbally pronounces them through a built-in speaker.

This interactive feature allows users to learn and familiarize themselves with Braille alphabets by listening to their pronunciation.



Mode II:

Mode 2 is an engaging quiz mode within our Braille literacy teaching device.

By connecting to a Bluetooth application, the device generates random Braille alphabets, challenging the user to correctly pronounce them using the app.

The device then evaluates the user's pronunciation and provides real-time feedback, fostering a dynamic learning experience.

With an Auto-connecting system.



BRAILLE.IO

Quiz

Mode III:

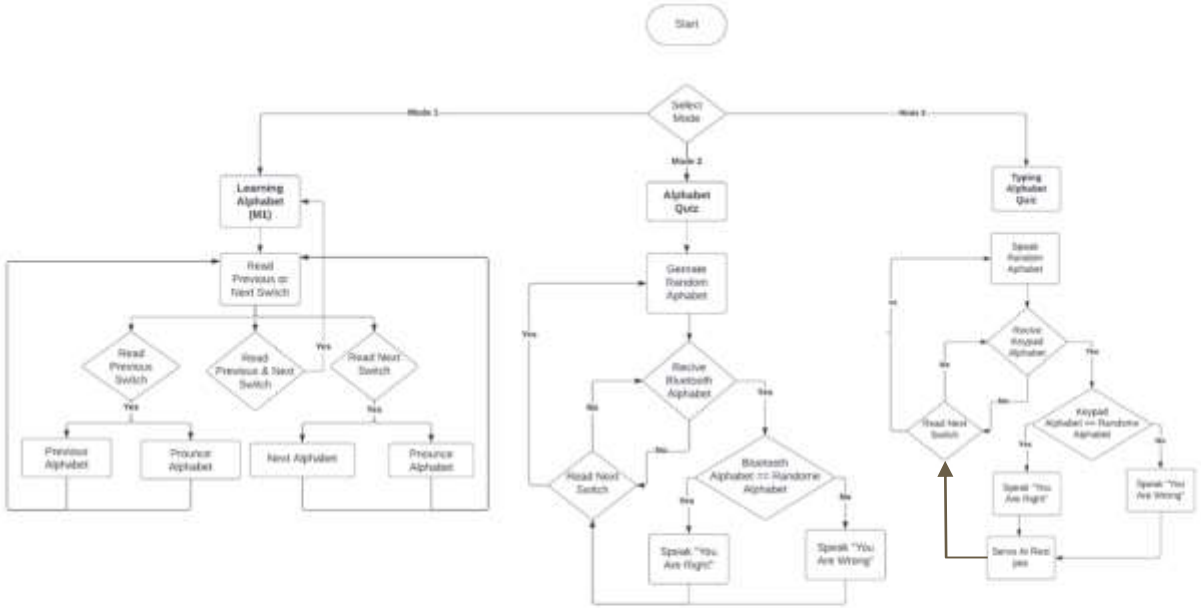
In this mode, users can practice Braille typing and pronunciation with real-time feedback.

It generates random Braille characters for users to type using a Braille keypad and provides immediate audio feedback.

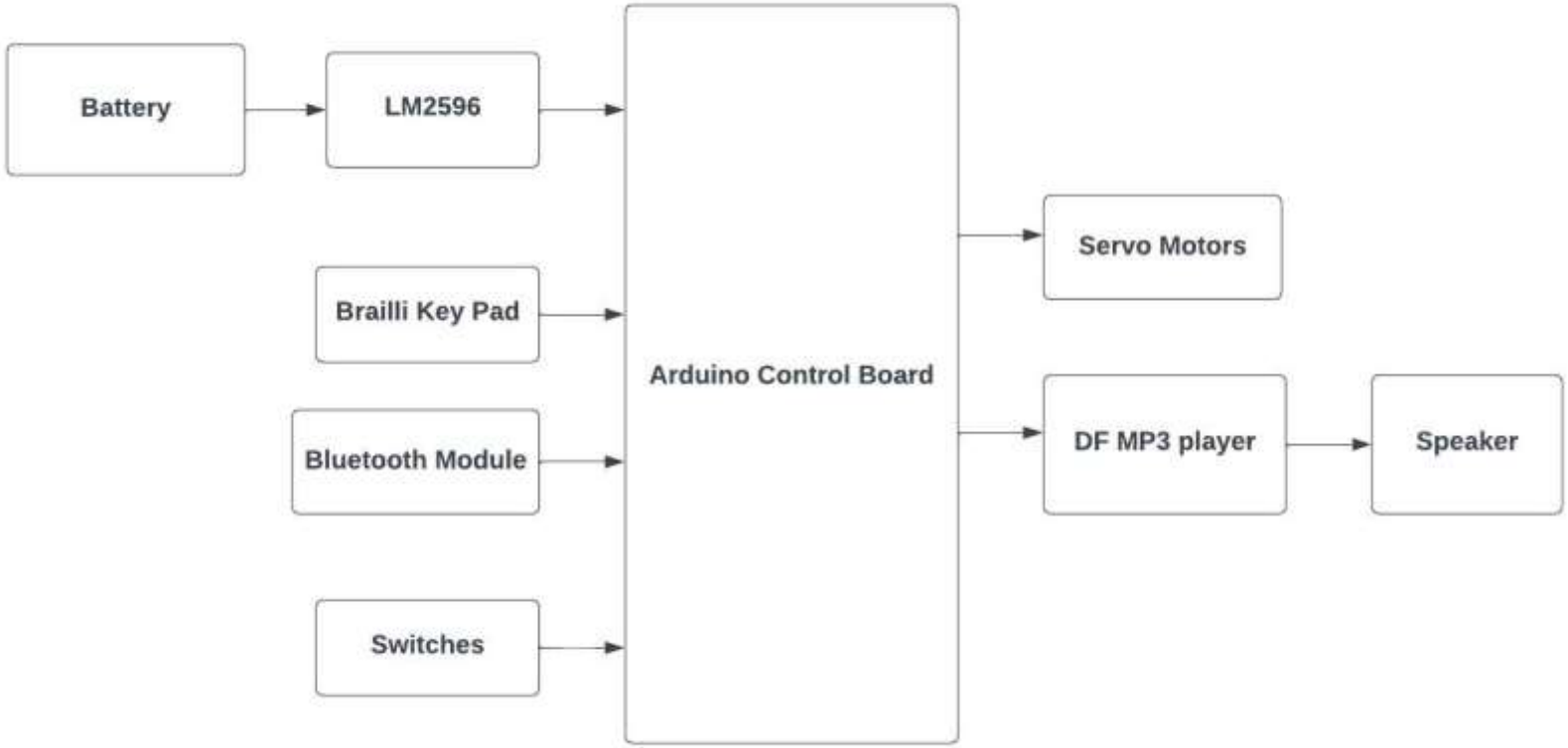
This mode aims to enhance Braille literacy, boost confidence, and improve practical Braille use for communication and writing.



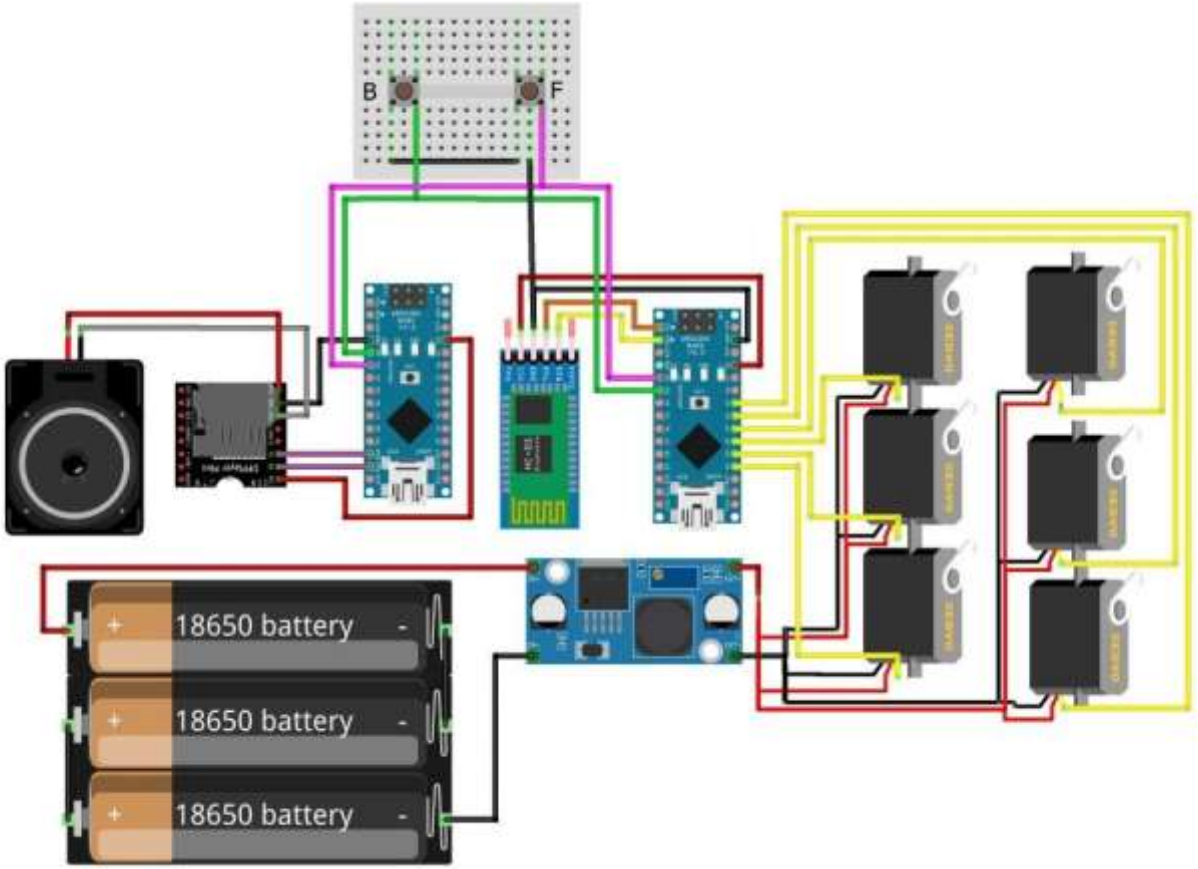
Flow Chart:


Start
Select Mode
Mode 1
Mode 2
Mode 3
Learning Alphabet (ML)
Alphabet Quiz
Typing Alphabet Quiz
Read Previous or Next Switch
Read Previous Switch
Read Previous & Next Switch
Read Next Switch
Yes
Previous Alphabet
Pronunce Alphabet
Next Alphabet
Pronunce Alphabet
Guessas Random Alphabet
Recive Bluetooth Alphabet
Read Next Switch
Bluetooth Alphabet == Randome Alphabet
Speak "You Are Right"
Speak "You Are Wrong"
Speak Random Alphabet
Recive Keypad Alphabet
Read Next Switch
Keypad Alphabet == Randome Alphabet
Speak "You Are Right"
Speak "You Are Wrong"
Yes
No

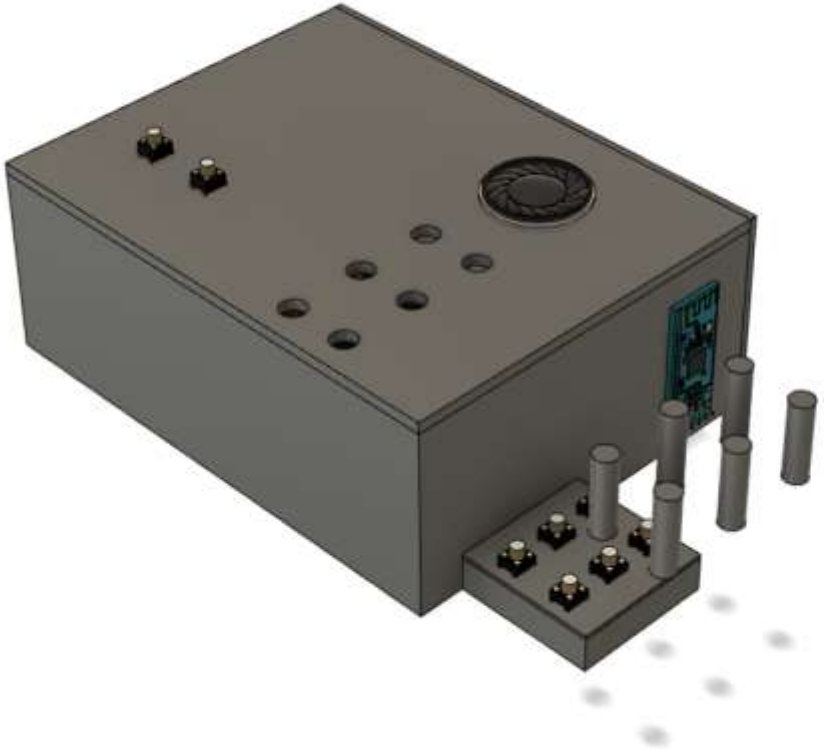
Block Diagram:



Circuit Diagram:



3D Design



Objectives:

Objectives:

- To Make Learning Easier
- To Help Visually Impaired people learn on Their Own
- To Use Technology for Education
- To Make Braille Fun and Engaging

Thank You